9. DFT and FFT Algorithms

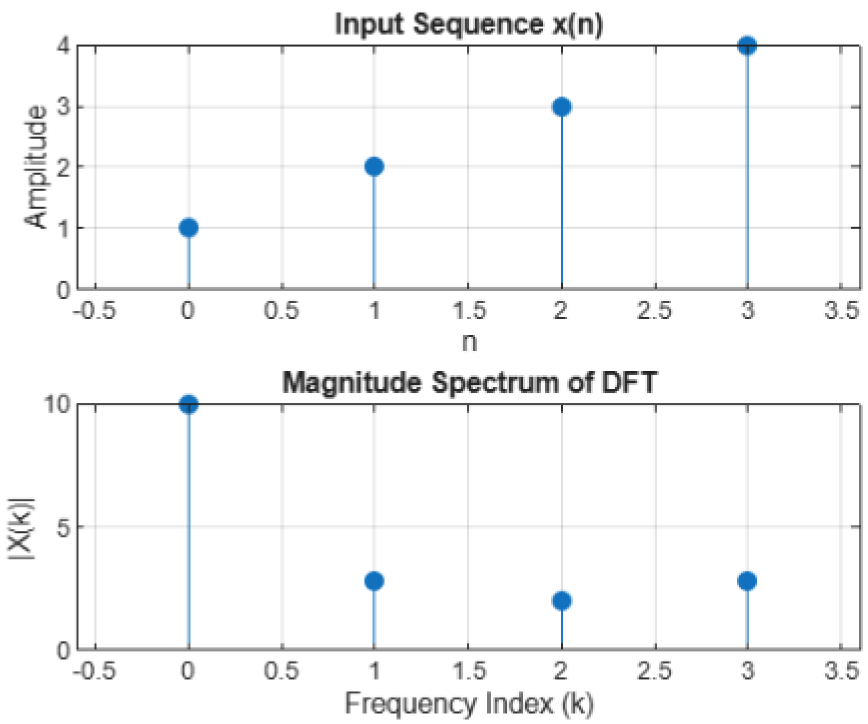
Experiment 9.1% Discrete Fourier Transform (DFT)

```
clc;
clear;
close all;
% Input sequence
x = [1 2 3 4];
% Length of sequence
N = length(x);
% DFT computation
X = zeros(1,N);
for k = 0:N-1
for n = 0:N-1
X(k+1) = X(k+1) + x(n+1)*exp(-1j*2*pi*k*n/N);
end
end
% Display DFT values
disp('DFT Output:');
disp(X);
% Plotting
figure;
subplot(2,1,1);
```

```
stem(0:N-1,x,'filled');
grid on;
title('Input Sequence x(n)');
xlabel('n');
ylabel('Amplitude');
subplot(2,1,2);
stem(0:N-1,abs(X),'filled');
grid on;
title('Magnitude Spectrum of DFT');
xlabel('Frequency Index (k)');
```

ylabel('|X(k)|');

```
DFT Output:
   10.0000 + 0.0000i  -2.0000 + 2.0000i  -2.0000 - 0.0000i  -2.0000 -
```



Experiment

9.2% Circular Convolution

clc;

clear;

close all;

% Input sequences

x1 = [1 2 3 4];

x2 = [1 1 1 1];

% Length

N = max(length(x1),length(x2));

% DFT of sequences

X1 = fft(x1,N);

X2 = fft(x2,N);

% Circular convolution

y = ifft(X1.*X2);

% Display result

```
disp('Circular Convolution Output:');
disp(y);
% Plotting
figure;
subplot(3,1,1);
stem(0:N-1,x1,'filled');
grid on;
title('Sequence x_1(n)');
xlabel('n');
ylabel('Amplitude');
subplot(3,1,2);
stem(0:N-1,x2,'filled');
grid on;
title('Sequence x_2(n)');
xlabel('n');
ylabel('Amplitude');
subplot(3,1,3);
stem(0:N-1,real(y),'filled');
grid on;
title('Circular Convolution Output');
xlabel('n');
ylabel('Amplitude');
```

Output

Input Sequences

x1(n)={1,2,3,4}, x2(n)={1,1,1,1}

Circular Convolution Result

y(n)={10,10,10,10}

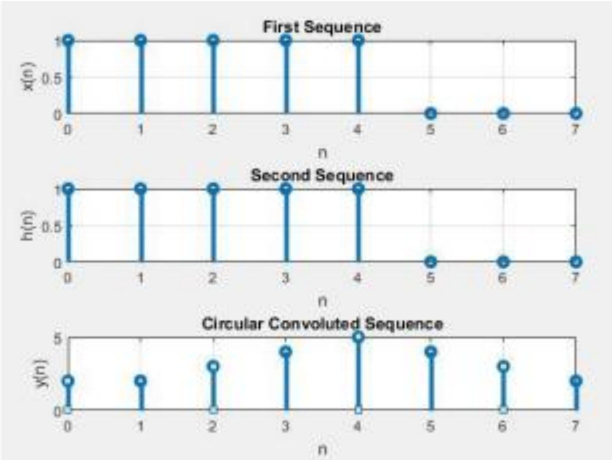
MATLAB Output

n y(n)

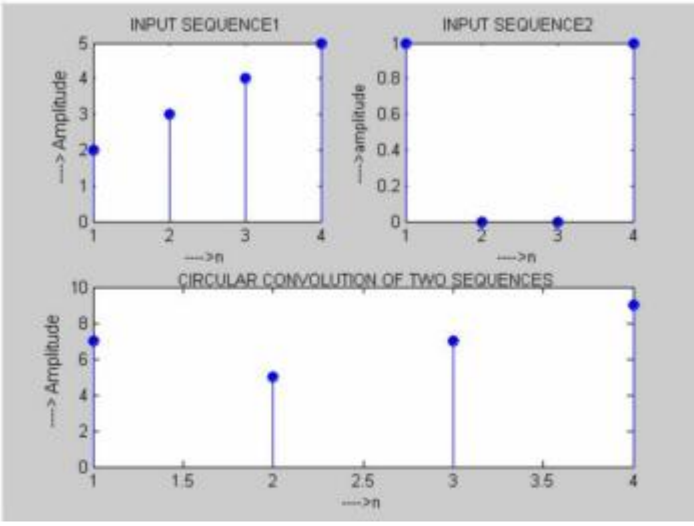
0 10

1 10

2 10



3 10



Experiment 9.3% FFT - Decimation In Time (DIT) Algorithm

clc;

clear;

close all;

% Input sequence

x = [1 2 3 4 5 6 7 8];

% FFT computation

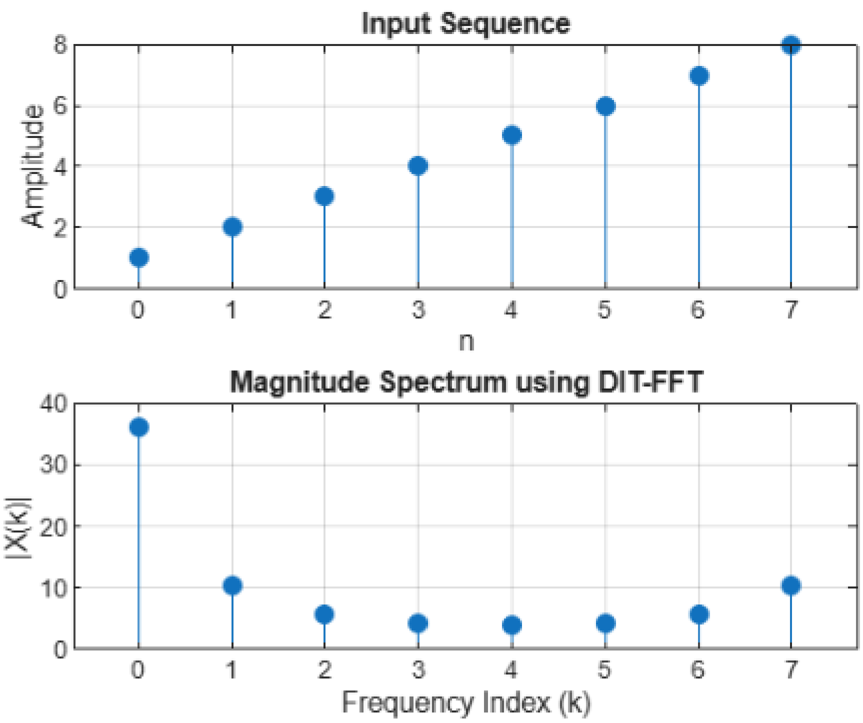
X = fft(x);

```
% Sequence length
N = length(x);
% Frequency index
k = 0:N-1;
% Display FFT values
disp('DIT-FFT Output');
disp(X);
% Plotting
figure;
subplot(2,1,1);
stem(0:N-1,x,'filled');
grid on;
title('Input Sequence');
xlabel('n');
ylabel('Amplitude');
subplot(2,1,2);
stem(k,abs(X),'filled');
grid on;
title('Magnitude Spectrum using DIT-FFT');

xlabel('Frequency Index (k)');
```

ylabel('|X(k)|');

```
DIT-FFT Output
  36.0000 + 0.0000i  -4.0000 + 9.6569i  -4.0000 + 4.0000i  -4.0000
```



% FFT - Decimation In Frequency (DIF) Algorithm

clc;

clear;

close all;

% Input sequence

x = [1 2 3 4 5 6 7 8];

% FFT computation

X = fft(x);

% Sequence length

N = length(x);

% Frequency index

k = 0:N-1;

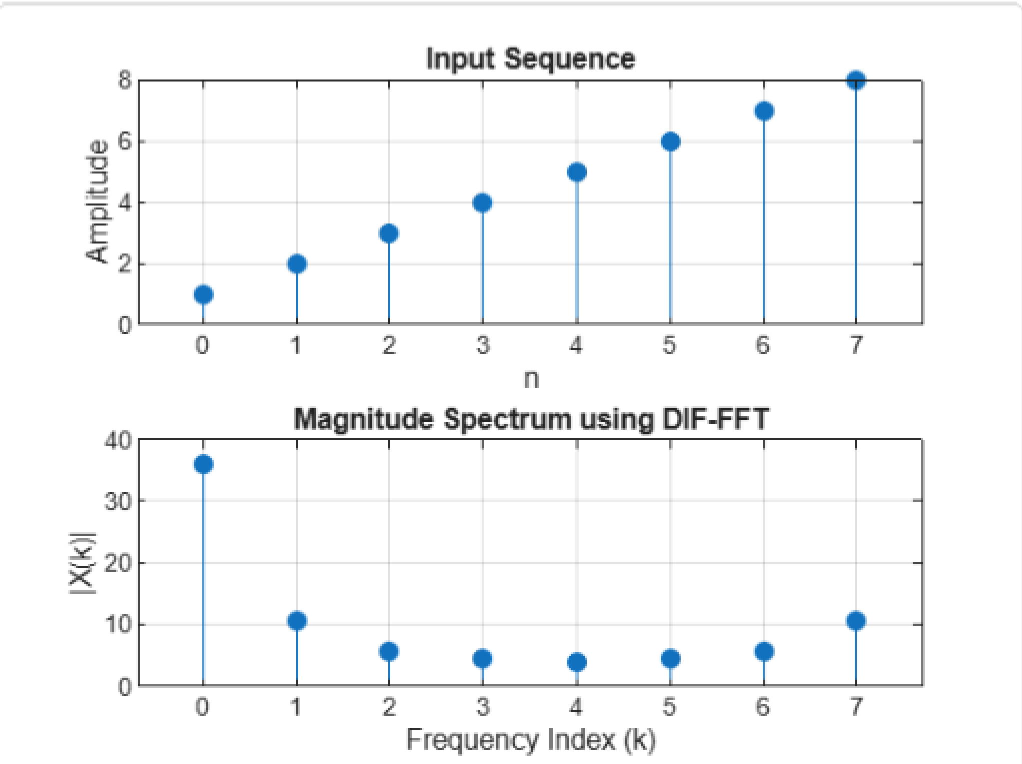
% Display FFT values

disp('DIF-FFT Output');

disp(X);

```
% Plotting
figure;
subplot(2,1,1);
stem(0:N-1,x,'filled');
grid on;
title('Input Sequence');
xlabel('n');
ylabel('Amplitude');
subplot(2,1,2);
stem(k,abs(X),'filled');
grid on;
title('Magnitude Spectrum using DIF-FFT');
xlabel('Frequency Index (k)');
ylabel('|X(k)|');
```

```
DIF-FFT Output
  36.0000 + 0.0000i  -4.0000 + 9.6569i  -4.0000 + 4.0000i  -4.0000 +
```



Experiment 9.4